



Cyberscope

A *TAC Security* Company

Audit Report

VaultedMetrics

June 2026

Network Ethereum

Address 0x76371561F18f3f5B4ed85a251ac9b245864b8a2A

Audited by © cyberscope

Table of Contents

Table of Contents	1
Risk Classification	2
Review	3
Audit Updates	3
Source Files	3
Overview	4
Findings Breakdown	6
Diagnostics	7
CCR - Contract Centralization Risk	8
Description	8
Recommendation	8
Team Update	8
OCTD - Transfers Contract's Tokens	9
Description	9
Recommendation	9
Team Update	10
Functions Analysis	11
Summary	13
Disclaimer	14
About Cyberscope	15

Risk Classification

The criticality of findings in Cyberscope's smart contract audits is determined by evaluating multiple variables. The two primary variables are:

1. **Likelihood of Exploitation:** This considers how easily an attack can be executed, including the economic feasibility for an attacker.
2. **Impact of Exploitation:** This assesses the potential consequences of an attack, particularly in terms of the loss of funds or disruption to the contract's functionality.

Based on these variables, findings are categorized into the following severity levels:

1. **Critical:** Indicates a vulnerability that is both highly likely to be exploited and can result in significant fund loss or severe disruption. Immediate action is required to address these issues.
2. **Medium:** Refers to vulnerabilities that are either less likely to be exploited or would have a moderate impact if exploited. These issues should be addressed in due course to ensure overall contract security.
3. **Minor:** Involves vulnerabilities that are unlikely to be exploited and would have a minor impact. These findings should still be considered for resolution to maintain best practices in security.
4. **Informative:** Points out potential improvements or informational notes that do not pose an immediate risk. Addressing these can enhance the overall quality and robustness of the contract.

Severity	Likelihood / Impact of Exploitation
● Critical	Highly Likely / High Impact
● Medium	Less Likely / High Impact or Highly Likely/ Lower Impact
● Minor / Informative	Unlikely / Low to no Impact

Review

Contract Name	VaultedMetricsReserve
Explorer	https://etherscan.io/address/0x76371561F18f3f5B4ed85a251ac9b245864b8a2A

Audit Updates

Initial Audit	27 Mar 2026
Corrected Phase 2	03 Apr 2026
Corrected Phase 3	08 Jul 2026

Source Files

Filename	SHA256
VaultedMetricsVM888ReserveCyberScope3.sol	8fdf2b2d901d4f52a373fd1d0ee835d4765 4aef09b5c822cec663567a72093aa

Overview

The VaultedMetrics Reserve contract implements an autonomous treasury and buyback mechanism for the VM888 ecosystem. The Reserve is designed to periodically utilize ETH reserves to execute strategic market operations without requiring continuous administrative intervention. Once the initial EMA configuration is completed and ownership is renounced, the contract is intended to operate through permissionless execution by external schedulers, keepers, or any public caller.

The Reserve organizes its operation into three execution chambers:

Floor Chamber, which purchases VM888 tokens, provides liquidity to the VM888/ETH liquidity pool, and permanently burns the resulting LP tokens to strengthen protocol-owned liquidity.

Reward Chamber, which acquires VM888 tokens and alternates their distribution between the staking and ecosystem wallets.

Shield Chamber, which accumulates VM888 tokens during normal market conditions and selectively deploys them as liquidity during significant market downturns or when predefined reserve thresholds are exceeded.

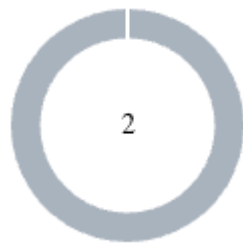
To determine its buying strategy, the contract maintains an Exponential Moving Average (EMA) derived from a Time-Weighted Average Price (TWAP) obtained from the Uniswap V2 pair. The EMA is used to classify market conditions as rising, stable, falling, or crashing, and to dynamically adjust buy sizes while bounding execution prices to reduce the impact of short-term market volatility and price manipulation attempts.

Reserve funding is intentionally restricted through a whitelist-based deposit mechanism that permits deposits only from predefined treasury addresses, while normal ETH transfers are rejected except for legitimate router refunds generated during liquidity operations.

As part of its design philosophy, the Reserve exposes permissionless execution functions, allowing anyone to trigger eligible Reserve operations or refresh the on-chain price oracle when appropriate. ETH usage is limited to the Reserve's VM888 buyback, reward, shield, and liquidity operations.

This design improves decentralization and ensures that autonomous operation does not depend on a trusted executor. Execution timing remains publicly observable, which is an inherent characteristic of permissionless AMM-based protocols. The contract mitigates the associated economic risks through TWAP-based pricing, EMA-bounded execution, spot-price validation, minimum output constraints, and liquidity protection parameters, while acknowledging the residual possibility of bounded MEV opportunities as a trade-off in favor of decentralized and permissionless operation.

Findings Breakdown



● Critical	0
● Medium	0
● Minor / Informative	2

Severity	Unresolved	Acknowledged	Resolved	Other
● Critical	0	0	0	0
● Medium	0	0	0	0
● Minor / Informative	0	2	0	0

Diagnostics

● Critical ● Medium ● Minor / Informative

Severity	Code	Description	Status
●	CCR	Contract Centralization Risk	Acknowledged
●	OCTD	Transfers Contract's Tokens	Acknowledged

CCR - Contract Centralization Risk

Criticality	Minor / Informative
Location	VaultedMetricsVM888ReserveCyberScope3.sol#L255,264,274
Status	Acknowledged

Description

The contract's functionality and behavior are heavily dependent on external parameters or configurations. While external configuration can offer flexibility, it also poses several centralization risks that warrant attention. Centralization risks arising from the dependence on external configuration include Single Point of Control, Vulnerability to Attacks, Operational Delays, Trust Dependencies, and Decentralization Erosion.

```
function setInitialEMA(uint256 initialPrice) external onlyOwner { (...)
function emergencyWithdrawETH() external onlyOwner nonReentrant { (...)
function emergencyWithdrawTokens(address token) external onlyOwner
nonReentrant { (...)
```

Recommendation

To address this finding and mitigate centralization risks, it is recommended to evaluate the feasibility of migrating critical configurations and functionality into the contract's codebase itself. This approach would reduce external dependencies and enhance the contract's self-sufficiency. It is essential to carefully weigh the trade-offs between external configuration flexibility and the risks associated with centralization.

Team Update

The team has acknowledged that this is not a security issue and states:

All onlyOwner functions (setChainlinkConfig, setInitialEMA, emergencyWithdrawETH, emergencyWithdrawTokens) are permanently disabled after renounceOwnership(). This is the same pattern used in the VM888 token contract, audited by Cyberscope with zero critical and zero medium findings after re-audit

OCTD - Transfers Contract's Tokens

Criticality	Minor / Informative
Location	VaultedMetricsVM888ReserveCyberScope3.sol#L264,274
Status	Acknowledged

Description

The contract owner has the authority to claim all the balance of the contract. The owner may take advantage of it by calling the `emergencyWithdrawETH` and `emergencyWithdrawTokens` functions.

Additionally due to the restrictions of the `_processDeposit` function it may be impossible to re-add the eth withdrawn.

```
function emergencyWithdrawETH() external onlyOwner nonReentrant {
    uint256 balance = address(this).balance;
    require(balance > 0, "No ETH to withdraw");
    (bool success, ) = msg.sender.call{value: balance}("");
    require(success, "ETH transfer failed");
    emit EmergencyWithdraw(address(0), balance);
}

function emergencyWithdrawTokens(address token) external onlyOwner
nonReentrant {
    uint256 balance = IERC20(token).balanceOf(address(this));
    require(balance > 0, "No tokens to withdraw");
    IERC20(token).safeTransfer(msg.sender, balance);
    if (token == VM888_TOKEN) {
        shieldBalance = 0;
    }
    emit EmergencyWithdraw(token, balance);
}
```

Recommendation

The team should carefully manage the private keys of the owner's account. We strongly recommend a powerful security mechanism that will prevent a single user from accessing the contract admin functions.

Temporary Solutions:

These measurements do not decrease the severity of the finding

- Introduce a time-locker mechanism with a reasonable delay.
- Introduce a multi-signature wallet so that many addresses will confirm the action.
- Introduce a governance model where users will vote about the actions.

Permanent Solution:

- Renouncing the ownership, which will eliminate the threats but it is non-reversible.

Team Update

The team has acknowledged that this is not a security issue and states:

All onlyOwner functions (setChainlinkConfig, setInitialEMA, emergencyWithdrawETH, emergencyWithdrawTokens) are permanently disabled after renounceOwnership(). This is the same pattern used in the VM888 token contract, audited by Cyberscope with zero critical and zero medium findings after re-audit

Functions Analysis

Contract	Type	Bases		
	Function Name	Visibility	Mutability	Modifiers
VaultedMetrics Reserve	Implementation	Ownable, ReentrancyGuard, AutomationCompatibleInterface		
		Public	✓	Ownable
	_ensureRouterApproval	Internal	✓	
	setInitialEMA	External	✓	onlyOwner
	emergencyWithdrawETH	External	✓	onlyOwner nonReentrant
	emergencyWithdrawTokens	External	✓	onlyOwner nonReentrant
	renounceOwnership	Public	✓	onlyOwner
		External	Payable	-
	deposit	External	Payable	-
	_processDeposit	Internal	✓	
	checkUpkeep	External		-
	performUpkeep	External	✓	-
	execute	External	✓	-
	_executeBatchBeat	Internal	✓	nonReentrant
	_executeFloor	Internal	✓	
	_executeReward	Internal	✓	
	_executeShield	Internal	✓	

	_deployShieldAsLP	Internal	✓	
	_getProportionalTokenAmount	Internal		
	_swapETHForVM888	Internal	✓	
	_getExpectedOutput	Internal		
	_requireSpotNearReference	Internal		
	_getCurrentPrice	Internal		
	_sanitizePriceAgainstEMA	Internal		
	_currentCumulativePrice	Internal		
	_observeTwap	Internal	✓	
	_getBuyAmount	Internal		
	_updateEMA	Internal	✓	
	_checkAndRefuelLink	Internal	✓	
	_getOracleExpectedLinkOutput	Internal		
	getReserveStatus	External		-
	getMarketCondition	External		-
	timeUntilNextBeat	External		-
	getCurrentPrice	External		-
	getAvailableETH	External		-

Summary

VaultedMetrics is an interesting project that has a friendly and growing community. The Smart Contract analysis reported no compiler error or critical issues. This audit investigates security issues, business logic concerns and potential improvements.

Disclaimer

The information provided in this report does not constitute investment, financial or trading advice and you should not treat any of the document's content as such. This report may not be transmitted, disclosed, referred to or relied upon by any person for any purposes nor may copies be delivered to any other person other than the Company without Cyberscope's prior written consent. This report is not nor should be considered an "endorsement" or "disapproval" of any particular project or team. This report is not nor should be regarded as an indication of the economics or value of any "product" or "asset" created by any team or project that contracts Cyberscope to perform a security assessment. This document does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors' business, business model or legal compliance. This report should not be used in any way to make decisions around investment or involvement with any particular project. This report represents an extensive assessment process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Cyberscope's position is that each company and individual are responsible for their own due diligence and continuous security. Cyberscope's goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies and in no way claims any guarantee of security or functionality of the technology we agree to analyze. The assessment services provided by Cyberscope are subject to dependencies and are under continuing development. You agree that your access and/or use including but not limited to any services reports and materials will be at your sole risk on an as-is where-is and as-available basis. Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives and other unpredictable results. The services may access and depend upon multiple layers of third parties.

About Cyberscope

Cyberscope is a TAC blockchain cybersecurity company that was founded with the vision to make web3.0 a safer place for investors and developers. Since its launch, it has worked with thousands of projects and is estimated to have secured tens of millions of investors' funds.

Cyberscope is one of the leading smart contract audit firms in the crypto space and has built a high-profile network of clients and partners.



A **TAC Security** Company

The Cyberscope team

cyberscope.io